

# PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

## Practical File Submission — Academic Session

Experiment No.: 03 | Course: PARS Lab | Framework: Python 3.14 / Flask | Repository: [dSAxmonis/pars-lab](#)

### Practical Name

#### Multimodal Feature Extraction from Audio Across Multiple Timestamps Using Flask

### AIM

To prepare an informative web page using Flask in Python to extract features from an audio recording across at least 10 timestamps, extracting at least 3 distinct features (RMS Energy, Zero Crossing Rate, and Spectral Centroid), justify their objectives for real-world applications, and embed the playable audio directly into the page.

### THEORY

#### 1. Temporal Short-Time Windowing

Audio waveforms are non-stationary over extended temporal sequences. To observe localized frequency and energy characteristics, short-time windowing ( $N = 50 \text{ ms}$ ) is applied around specific timestamps  $t_i$ :

$$x_{\{t_i\}}[n] = x[n] \times w[n - n_i]$$

#### 2. Extracted Audio Features & Mathematical Formulas

##### 1. Root Mean Square (RMS) Energy:

$$\text{RMS}_t = \sqrt{(1 / N) * \sum [x_t[n]^2]}$$

*Application Objective:* Voice Activity Detection (VAD), silence removal, loudness metering, and acoustic event onset detection.

##### 2. Zero Crossing Rate (ZCR):

$$\text{ZCR}_t = (1 / 2N) * \sum | \text{sgn}(x_t[n]) - \text{sgn}(x_t[n-1]) |$$

*Application Objective:* Distinguishing voiced harmonic speech (vowels) from unvoiced fricatives (/s/, /sh/) and percussion noise.

##### 3. Spectral Centroid (Frequency Center of Mass):

$$C_t = \sum [f_k * |X_t[k]|] / \sum |X_t[k]|$$

*Application Objective:* Acoustic timbre classification, sound brightness evaluation, musical genre recognition, and speaker profiling.

## STEPS & EXECUTION

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### Step 1: Environment Setup & Dependency Verification

Configured `exp 3/` containing `app.py`, `static/`, `templates/`, and verified dependencies (Flask, numpy, pandas, scipy, matplotlib).

### Step 2: Audio Ingestion & Signal Properties

Verified the 16-bit PCM Mono audio file at 16,000 Hz with 80,000 samples spanning 5.0 seconds.

### Step 3: Multi-Timestamp Feature Extraction Execution

Executed extraction logic across 10 timestamps ( $t = 0.5s, 1.0s, \dots, 5.0s$ ) calculating RMS Energy, ZCR, and Spectral Centroid.

### Step 4: Web Application Launch on Port 5005

Launched the Flask web server on <http://127.0.0.1:5005> to render the interactive dashboard with playable audio.

# OUTPUT / DASHBOARD RESULTS

## Output 1: Embedded Audio Player & Signal Specifications

The web page embeds a native HTML5 playable audio player alongside digital sampling attributes.

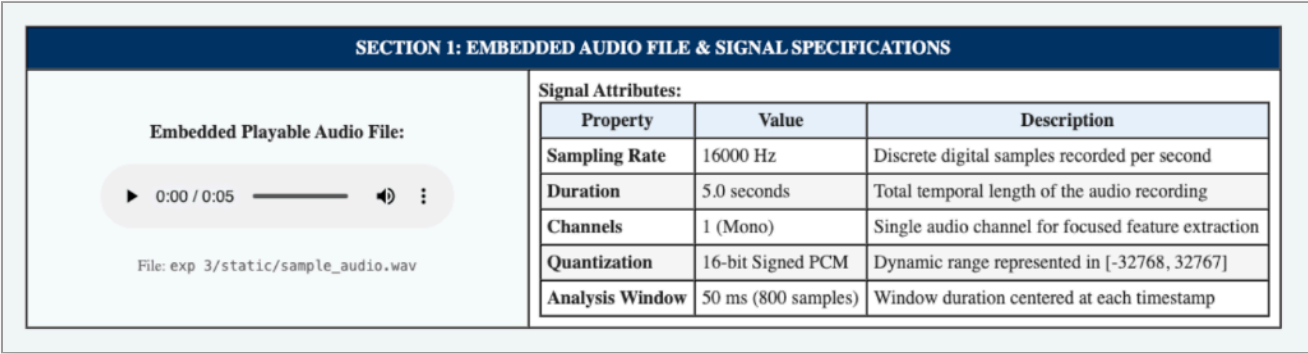


Figure 3.1: Section 1 — Playable audio player and signal metadata table.

## Output 2: Feature Extraction Across 10 Timestamps

Extracted numerical features (RMS Energy, ZCR, Spectral Centroid) for all 10 timestamp samples.

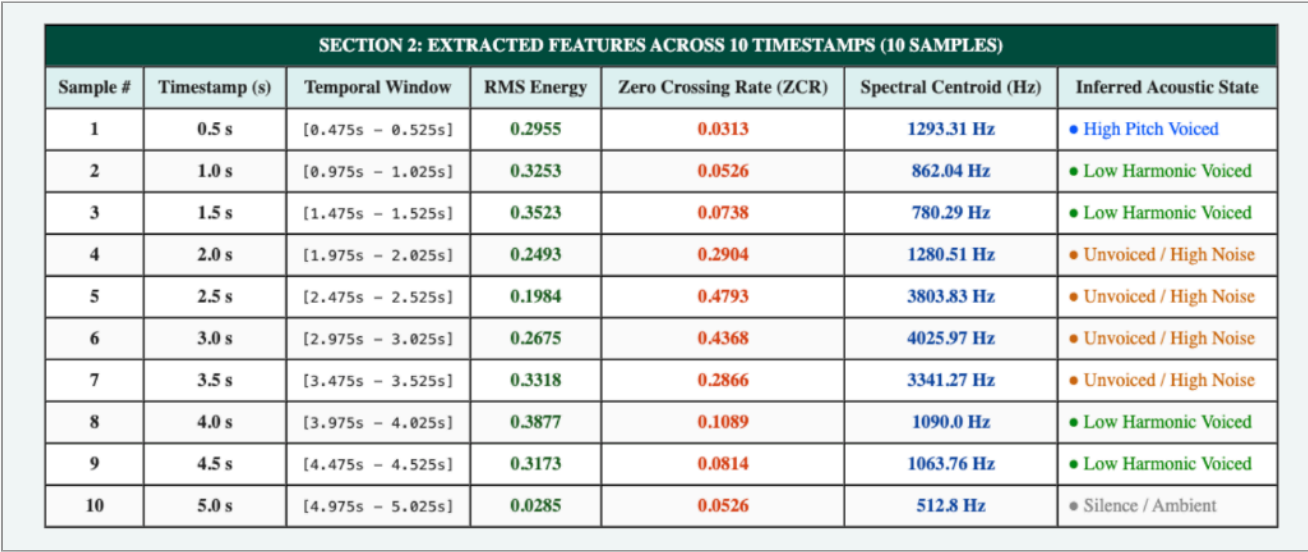


Figure 3.2: Section 2 — 10 timestamps feature extraction table with inferred acoustic state.

Output 3: Temporal Feature Trajectory Multi-Plot

Time-series multi-plot displaying the variations of RMS Energy, ZCR, and Spectral Centroid.

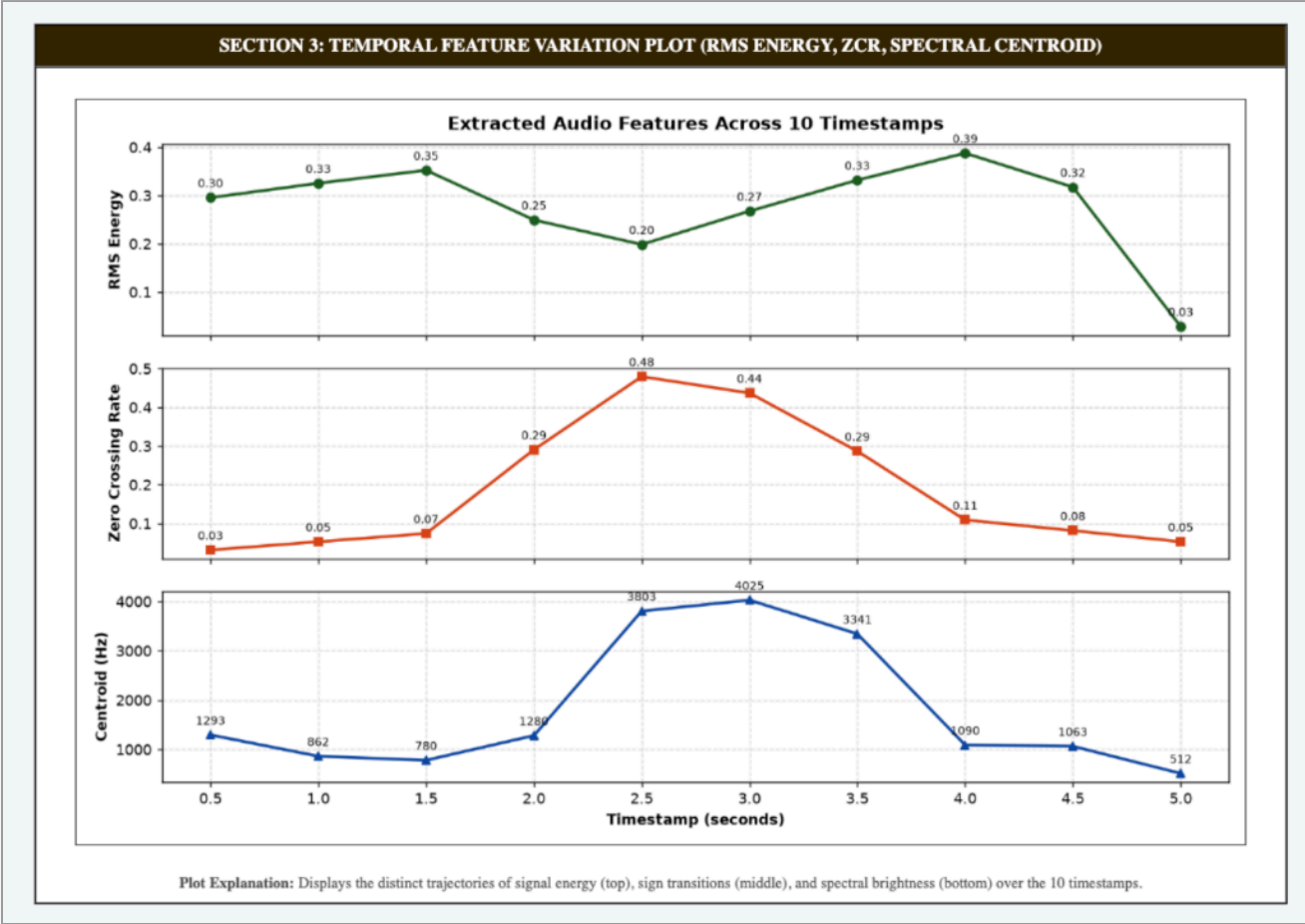


Figure 3.3: Section 3 — Multi-panel Matplotlib plot across timestamps.

Output 4: Application Objective Justification Table

Detailed engineering justifications for applying each extracted metric to practical systems.

| SECTION 4: JUSTIFICATION OF EXTRACTED FEATURES FOR ENGINEERING APPLICATIONS |                                                            |                                                                                                |                                                                                                                         |
|-----------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Extracted Feature                                                           | Mathematical Formula                                       | Acoustic / Physical Interpretation                                                             | Application Objective Justification                                                                                     |
| RMS Energy (Root Mean Square)                                               | $\sqrt{(1/N) * \sum(x[n]^2)}$                              | Measures instantaneous physical signal power, loudness, and amplitude.                         | Voice Activity Detection (VAD), silence removal in speech systems, audio segment boundary identification.               |
| Zero Crossing Rate (ZCR)                                                    | $(1 / 2N) * \sum( \text{sgn}(x[n]) - \text{sgn}(x[n-1]) )$ | Measures the rate of sign changes between successive discrete audio samples.                   | Distinguishing voiced speech (low ZCR vowels) from unvoiced speech (high ZCR fricatives like /s/, /sh/) and percussion. |
| Spectral Centroid (Frequency Center)                                        | $\sum(f_k *  X[k] ) / \sum( X[k] )$                        | Indicates the center of gravity of the frequency spectrum; corresponds to acoustic brightness. | Musical timbre classification, audio genre recognition, instrument identification, speaker voice quality profiling.     |

Figure 3.4: Section 4 — Application objectives and mathematical definitions.

## CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

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### GitHub Repository (Clickable):

GitHub: dSAxmonis/pars-lab

URL: <https://github.com/dSAxmonis/pars-lab>

### Direct Source Code Files:

- **Flask Server Application:**  
[https://github.com/dSAxmonis/pars-lab/blob/main/exp\\_3/app.py](https://github.com/dSAxmonis/pars-lab/blob/main/exp_3/app.py)
- **HTML Template:**  
[https://github.com/dSAxmonis/pars-lab/blob/main/exp\\_3/templates/index.html](https://github.com/dSAxmonis/pars-lab/blob/main/exp_3/templates/index.html)
- **Dependencies:**  
[https://github.com/dSAxmonis/pars-lab/blob/main/exp\\_3/requirements.txt](https://github.com/dSAxmonis/pars-lab/blob/main/exp_3/requirements.txt)
- **Lab Manual (Markdown):**  
[https://github.com/dSAxmonis/pars-lab/blob/main/exp\\_3/LAB\\_PRACTICAL\\_FILE.md](https://github.com/dSAxmonis/pars-lab/blob/main/exp_3/LAB_PRACTICAL_FILE.md)

## CONCLUSION

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In this experiment, an informative web application was developed using Flask in Python to perform temporal feature extraction on an audio signal. Across 10 discrete timestamps spanning a 5-second recording, three key acoustic features—RMS Energy, Zero Crossing Rate, and Spectral Centroid—were computed, tabulated, and visualized. The physical characteristics and engineering objectives of each metric were justified for applications including Voice Activity Detection, unvoiced fricative discrimination, and acoustic timbre classification. Finally, the playable audio source was embedded into the dashboard.